

The Elephant in a Neuron

Metrological certification of the continuous custody of a closed volume between two stations

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EVIDENTIARY STATUS NOTICE

No figure in this document comes from a bench measurement of our own. Every value is a derivation, a simulation or a citation, and carries its declared status (MEAS · DER · LIT · DEC). Across the framework's eight axes of completeness, the seven statement axes stand at 100 % and the eighth — MEASURED — stands at **0 %**. The framework is at **87.5 %**.

This document is a **pre-registered protocol for third parties to refute**, not a report of results.

Typographic note: the Emilio unit is set here as ■; no system typeface carries its own glyph (U+2C4E).

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Abstract

We set out a complete, costed and falsifiable theoretical framework for the physical transport of a **closed volume** between two stations, together with a metrological protocol that certifies that what arrives is what departed. The declared architecture is the **complete and physical** transport of a whole human inside a **bubble/object, without cutting and without reading the interior**: the interior is a cabin that is never modified, because the laws of nature maintain it; what is transported and certified is the **shell**, and the human travels inside as **cargo**.

Excluded from the first day: wormholes, speculative quantum formalism, and magical dematerialisation. Nothing here proposes a new mechanism of nature. What is proposed is **metrology, protocol, and evidence accounting**.

The framework rests on an agreement contract $A = \sqrt{(\Omega_{\text{imp}}/\Omega_{\text{act}})}$ evaluated against the **nearest** impostor, with an explicitly declared **dead zone**, and a false-acceptance rate reported with its own sample size under the rule of three. It yields a set of hard limits that were derived, not assumed: a **5.03 m ceiling** on bubble size, a **1 g** declared acceleration, an obligation that **every act be exo-atmospheric**, and an operating principle — *the door is never the punishment* — that separates delivery from certification.

Across eight axes of completeness, the seven **statement** axes stand at 100 % and the eighth — **MEASURED** — stands at **0 %**. The framework is at **87.5 %**, and the missing 12.5 points are a single axis costing **€1,674**.

This document is a pre-registered protocol for third parties to refute, not a report of results.

1. Genesis: why a neuron, and why an elephant

The programme began with a metaphor and survived it. The question was whether the information content of a nervous system could be transferred, and the elephant in the title is the absurdity of the scale involved: fitting something enormous through something very small.

That framing was wrong, and correcting it was the single most productive act of the whole programme.

1.1 The globe

A connectome is not a brain being measured. It is a GLOBE.

object in the graph	what it is
a node	a city — neuron A = London , neuron B = New York
a whole connectome	a country or a continent
the set of connectomes	the planet
an edge	which places can be reached from a place
the weight	coupling between places
the impostor	a wrong destination address — never "another individual"

We speak always of a **human body**, never of a *nervous system*. Measuring on worm or fly tissue is a **test bench for space**, not neuroscience: the biological substrate is the vehicle, not the subject.

And a **teleportation is measured between two PLACES IN THE SAME SPACE** — London → New York — never between two individuals. Comparing two distinct individuals **is not a teleportation**.

The vocabulary of "measuring brains" narrowed the design of experiments for weeks without any number demanding it. Under the space framing, one experiment showed that the act certifies equally well in human cardiac tissue as in

nervous tissue — band $8.97\times$ against $9.37\times$ — and the largest declared limit of the programme fell. It is an experiment that would never have been designed under the old frame.

The same correction had already been made once: **the synapse is the MECHANISM to emulate as a vehicle, not the object of study**. Same shape of error, same remedy.

1.2 The investigator's role, and what the collaboration actually was

This work was not produced by an assistant with a spectator. The declared architecture, the direction of the programme, the search for external evidence, and several of the methodological contributions are the investigator's, and without them the programme would have stalled against the walls the assistant kept hitting.

Specifically, the following are the investigator's own contributions:

- **The connectome-as-globe framing** (§1.1), which unblocked the experimental design.
- **The banking-transfer analogy** applied to the protocol, from which the two-phase commit, finality, reconciliation, four-eyes and the nostro/vostro ledger entered the framework — and with them the discovery that the act had no defined abort semantics.
- **Cataloguing, measuring and defining the LIMBO and its classes** (§5-ter, Annex A VI).
- **The creation of the MATE system** (§1.4).

The assistant contributed calculation, attack batteries, and drafting. The architecture decisions are the investigator's; the arithmetic errors, several of which are documented in this paper because they were caught and corrected, were the assistant's.

1.3 The closed-route episode

At one point the programme spent two days comparing incommensurable magnitudes, and at another it announced 110 attacks while having executed 38. Both were caught by the investigator asking a plain question. These episodes are recorded rather than hidden, because the method claims that **optimistic accounting is the characteristic failure mode of this kind of work**, and a framework that claims that must show its own instances.

1.4 TRIADA and MATE

TRIADA — before spending computation:

1. **INVENTORY** — is it already measured, is it already on disk?
2. **MATHEMATICS** — does a bound answer it without simulating? **Written BEFORE running**, with numerical predictions P1/P2/P3.
3. **MATE** — what computation can be reused?

MATE — *Methodology of Advance by Strategic Tension*, created by the investigator from the projection of a mate-in-three in chess. Five rules:

- **R1 · conservation of tension.** A line that releases the tension is abandoned even if it looks safe.
- **R2 · invariance of the rules.** The rules of the game do not change to suit the plan.
- **R3 · projection of the last three.** Before moving, the final three moves are named.
- **R4 · advance through critical squares.** Progress is measured by squares that matter, not by moves made.
- **R5 · mate $\neq$ theatre.** A position that looks winning and is not, is a loss.

TRIADA is the opening and MATE is the endgame. TRIADA is R3 turned into procedure.

2. The phases of the programme

The programme passed through the following stages, each of which ended when a number, not an opinion, closed it.

Phase 1 · the metaphor. Connectomics and the synapse as an information channel. Closed by the globe correction: the substrate is a vehicle.

Phase 2 · thermodynamic and informational limits. Bekenstein-type bounds on what a reading architecture would require. Closed by the decision to **not read**.

Phase 3 · non-cloning. Parfit's paradox and the biological consequences of any architecture that copies. Closed for the same reason: with $\phi = 1$ and no reading, the question does not arise.

Phase 4 · baryonic transit. The mass must travel. This is the point at which the programme stops being about information and starts being about metrology.

Phase 5 · the act of the object. Multi-body matter, the $3n - 6$ spatial law, and the minimum number of constraints needed to fix a configuration.

Phase 6 · closed-volume custody. The shell becomes the certified object and the interior becomes cargo. This is the declared architecture.

Phase 7 · the verification engine. The contract, the impostor, the floor, GATE-0, and common-mode cancellation.

Phase 8 · the protocol. The pipeline, the limbos, the abort semantics, the operating economics — the content of §5-bis through §5-quater.

A note on what was dropped. An earlier architecture assumed *destruction plus reconstruction*, forced by dose considerations. That architecture is **superseded**: it belonged to a frame in which the interior had to be read. With no reading, there is nothing to destroy and nothing to reconstruct.

3. The formal framework

The framework is a set of statements, each carrying its evidentiary status, its units, a falsifying experiment, and that experiment's price. The complete formulary is Annex A; what follows are the load-bearing results.

3.1 The contract

$\Omega = 1 - \text{corr}$	disagreement
$A = \sqrt{(\Omega_{\text{imp}} / \Omega_{\text{act}})}$	band, against the NEAREST impostor
$dA/A = \frac{1}{2} \sqrt{(u_{\text{imp}}^2 + u_{\text{act}}^2)}$	dead zone

The contract exists if and only if $A > 1 + dA$. Two points deserve emphasis because both were defects that had to be found and fixed:

- The impostor is the **nearest**, not the median. Taking the median of the impostor distribution flatters the band; the tail is what matters, exactly as in biometric practice.
- Without a dead zone, $A = 1.05$ was being reported as a contract. **It is a tie.**

Measured over $3 \cdot 10^6$ impostors: $\Omega_{\text{act}} = 8.16 \cdot 10^{-4}$, nearest impostor $4.99 \cdot 10^{-2}$, $A = 7.82$, $d' = 5.5$.

3.2 The false-acceptance rate, reported honestly

$$\text{FAR} \leq 3/k$$

rule of three (Hanley & Lippman-Hand, 1983)

A rate is never reported without its **k**. Over $3 \cdot 10^6$ impostors with zero acceptances, **FAR** $\leq 10^{-6}$. The same discipline applies to the operational failure rate: *zero deaths in 37* means **at most one in twelve**, not zero.

3.3 The act is material, not informational

$$\psi = g \cdot K^4 / A^4 = 1.03 \cdot 10^{-4}$$

The act is **9,713 times more material than informational**. This single number is what licenses the whole framework to be metrology rather than information theory.

3.4 The Non-Fusion Theorem

With $\phi = 1$ and no reading, the *Brundlefly* failure — two configurations mixing — **is impossible by construction**: there is no pattern to mix. This is not a safety measure. It is a consequence of refusing to read.

3.5 The law of the map: $v \geq L$

$$v \geq L$$

v/L converges to 0.6

v is the size of the OPERATION that formed the part. Point-by-point milling: 100–500 mm. Mould, press, deep-draw, hydroform: the whole part.

The two shells must be formed in ONE operation with the same tooling.

This eliminates welds as a failure mode, and it is the most falsifiable and most potentially novel claim in the framework. It is also, at €25, the cheapest thing in it to test.

3.6 GATE-0 — the veto the architecture hangs from

$$D(r) = C \cdot r^\alpha$$

structure function over paired fiducials

VETO if $\alpha < 0.35$

reliable at 98.7 %

PASS if $\alpha \geq 0.95$ AND $D(176 \text{ mm}) \leq 35 \mu\text{m}$

INTERMEDIATE otherwise

NOT sufficient to launch the act

$\alpha \approx 1$ means the residual error field is **smooth**, and a local affine kills it as $(r/L)^2$. $\alpha \approx 0$ means the error is **independent per site**, and the hierarchy of references does not work at all. One exponent answers *is this error correctable?*

Four defects had to be repaired here, and they are instructive:

- The amplitude criterion was originally a fixed threshold on C . Since α is a ****fitted** parameter, **the units of C change with every fit: the threshold was dimensionally invalid****, and it never once bit in 1,800 fields. It is now $D(r_0)$ in micrometres at a reference separation.
- α was estimated by ordinary least squares, whose breakdown point is $1/n$. A single $50 \mu\text{m}$ speck on 25 fiducials dragged α from 0.974 to 0.765 and turned PASS into INTERMEDIATE. The estimator is now **Theil-Sen** (breakdown 29.3 %).
- The fiducial plate was a **grid**, which is blind to the 40 error a three-axis mill

actually produces. It is now **five rings**.

- A **perfect** field produced $\alpha = \text{NaN}$ and a verdict of "NO DATA": the instrument failed in

the best possible case, which is the worst place to fail. A field below the reader's floor is not absence of data — it is a perfectly smooth field, and that is $\alpha = 1$ by construction.

The declaration that GATE-0 is a **reliable veto (98.7 %)** and a **poor pass (46 %)** is kept explicit, because a test that is honest about which of its two answers to trust is rarer than it should be.

3.7 The percentile that lies at your sample size

$$\text{PCT_INV} = 100 \cdot (1 - 1/2n)$$

A fixed percentile threshold misrepresents detection power at small n . Using p98 with $n = 50$ detected **10 %** of the cases where 90 % had been declared. The percentile must be tied to the sample size.

4. The phase diagram: where this framework applies

The framework is not universal and says so. Three boundaries delimit it.

By size. $\Omega \propto 1/L$ makes a larger cabin *easier* to certify, while α decreasing with L makes GATE-0 *impossible*. The two effects are opposed, and $v \geq L$ comes out of the crossing. The hard ceiling is computed in §5-ter.4: **$L \leq 5.03 \text{ m}$** .

By mass and energy. $t_{\text{wall}} \propto L$, $M_{\text{shell}} \propto L^3$, $E \propto M \propto L^3$. From 30 cm to 3 m the shell goes from 9 to 888 kg — **×99 of energy for the same trip in the same time**.

By distance. The transit-time law $t = 2\sqrt{(d/a)}$ is valid only while the peak velocity stays sub-escape. At 1 g this holds to **12,786 km**; at 9 g only to **1,421 km**. Beyond that, the profile is not a hop but a departure from Earth, and the law does not model the required coast phase.

5. THE TIME FACTOR · nothing is instantaneous, and the trip is rarely the longest part

5.1 Two separable terms

$$t_{\text{total}} = t_{\text{handling}(m)} + t_{\text{transit}(d)}$$

The first term depends on **mass** and on the certification work; the second on **distance**. They are separable, and the surprise is which dominates.

5.2 The three formulas of time

$$\begin{aligned} t_{\text{transit}} &= 2 \sqrt{(d / a)} & a &= g_{\text{sustained}} \cdot 9.81 \\ t_{\text{floor}} &= 10 \text{ re-placements, or } k = 3 \text{ control points if inherited} \\ t_{\text{act}} &= t_{\text{pipeline}} + t_{\text{transit}} \end{aligned}$$

Time is set by the OCCUPANT, not by the mass. Mass sets the **energy**, which scales as L^3 .

5.3 The crossing distance is a staircase

For a person, the crossing distance — where transit starts costing more than certifying — is **358,983 km**. The Moon is at 384,400 km. **The Moon falls almost exactly where the journey begins to cost more than certifying it.**

(Honesty note: this figure is sensitive to the handling model at the 10 % level. The conclusion holds either way because the Moon sits inside that margin, but `t_handling` is the term most worth measuring, and it is marked DEC rather than DER for that reason.)

5.4 What this section closes

Nothing is instantaneous. The time of a teleportation is not set by the destination: it is set by the mass and by the certificate, and the destination only adds a term which, below the Moon, is the minor one. An act is fast when the cargo fits in one hand and the destination is far. It is slow when a crane is needed, even if the destination is one kilometre away.

5-bis. THE PIPELINE · the 24 states of the act, and the branch nobody had written

Everything above describes **what** is certified. This section describes **how it is executed**: the complete sequence from powering on cabin A to the subject stepping out of cabin B. It is written as a state machine because that is how it would have to be programmed.

5-bis.1 The 24 states

	state	clock	what happens	verification
S0	STORED	—	unpowered, in the room at ± 1 K	soak = 0 at start-up
S1	START-UP	T-35	the cabin is powered on	self-test of the 13 instruments
S2	FUNCTIONAL VERIF.	T-33	scale, O ₂ and recorder against standards	the report is signed; its hash goes into the message
S3	RESERVATION	T-32	A proposes, B reserves or refuses	phase 1 of the two-phase commit
S4	WEIGHING	T-30	the occupant, dressed and with luggage	against the cabin's maximum mass
S5	BIOMETRIC REF.	T-28	reference iris capture	without it there is nothing to compare against
S6	INSPECTION	T-25	visual of the interior + signed declaration	against the stowaway
S7	BOARDING	T-20	enters, is restrained, the door closes	closing torque recorded
S8	SEALING	T-15	SHA-256 commitment + timestamp, two signatures	blinding + four eyes
S9	FLOOR A	T-0	10 re-placements, or $k = 3$ control points if inherited	Shewhart chart
S10	READ A	T+12	25 fiducials + 50 diameters	$R^2 \geq 0.98$
S11	SIGN AND SEND	T+14	1,806 B message, Ed25519 signed	B verifies before touching anything
S12	PRESSURISATION	T+16	ramp ≤ 13.5 psi/min	the CO₂ clock starts: 15.8 min
S13	COMMIT	T+18	B can no longer refuse	return possible up to 30 % of route
S14	TRANSIT	T+18	3.4 min at 9 g	null mitigation
S15	ARRIVAL	T+21	braking, at rest w.r.t. the destination floor	

	state	clock	what happens	verification
S16	VENTILATION B	T+22	opens	10 of the 15.8 CO ₂ minutes used
S17	FLOOR B	T+22	inherited, or 10 re-placements if first	
S18	READ B	T+34	the six criteria + FAR with its k	
S19	RECONCILIATION	T+36	A's record checked against B's	
S20	BLIND OPENING	T+38	the system opens with the sealed seed	no human decides
S21	EXIT BIOMETRICS	T+39	iris against the S5 reference	declared FRR
S22	DISEMBARKING	T+40	the occupant steps out	
S23	CLOSING	T+42	archived, cabin balance updated	

77 minutes of powered machine for 3.4 of flight.

5-bis.2 What the pipeline found

First: 9 of 17 abort branches had no destination. The framework had been attacked six times on the physical steps and **not once on the control flow**. The nine orphan branches all fall after sealing — that is, **with a person inside** — while the eight before it return to S0 and are trivial. Sealing was not merely the administrative point of no return: it was **the boundary between two different abort regimes**.

Second: the battery did not close. It had been sized for the act (3.4 min) and the pipeline lasts twenty times longer: $105 \text{ W} \times 77 \text{ min} = 135 \text{ Wh}$ **against 124 of battery**. The answer is not more battery — the cabin runs on an **umbilical** while in station, and the battery only covers S12→S16: 7.4 min, 13 Wh, $9.6\times$ margin. That forces a **hot-disconnect power penetration** that the penetration count had missed.

Third, and the one that matters: THE DOOR IS NEVER THE PUNISHMENT.

- **Before commit there is rollback.** The act did not happen; the person leaves the way they came in. Read A gains a **hard limit of three retries** — without a limit it was a loop with an occupant inside — and a pressure abort descends **at the same rate** at which it rose.

- **After commit there is no rollback.** Delivery is always fulfilled; the only thing that can fail is the certificate. The five failures from S15 to S21 all resolve **the same way** — **by opening**: arrival out of tolerance opens and does not certify; reconciliation that does not match opens and leaves the act **in dispute**; a sealed seed that fails to decrypt opens and leaves the act **null** — it does not contaminate the rate; failed exit biometrics opens and credentials manually, because locking someone in over a 10^{-3} FRR is unacceptable.

The framework had six criteria for denying a certificate and **none** for denying an opening, because denying an opening is not an option of the system. It was true, and that is why nobody had written it. **What is implicit cannot be audited.**

5-ter. WHAT THE FRAMEWORK LEARNED AFTER THE PIPELINE

5-ter.1 The acceleration was a decision, and it was the wrong one

The framework declared 9 g for hundreds of decisions without anyone asking why. Checking a table of real times revealed that $E = m \cdot a \cdot d$ is **linear in acceleration**: dropping to 1 g multiplies transit by 3 and **divides energy by 9**.

And transit does not matter. The act is **97 % paperwork**: London–Berlin goes from 3 to 10 minutes of flight, and from 2.0 to 2.1 hours door to door. In exchange, at 1 g the neck criterion disappears — an infant carries **25 %** of its mass in the head against an adult's 7 % — along with aspiration on vomiting, the anti-g suit, the reclined posture, loose objects as projectiles, and nearly the whole medical fitness list.

Seven of the thirteen occupant gaps did not come from the act: they came from the acceleration.

The reach of the transit law also multiplies by 9, since $v = \sqrt{(a \cdot d)}$: from 1,421 km to **12,786 km**. Declared: **1 g**.

5-ter.2 The contradiction hidden for 715 decisions

The framework demanded $D(176 \text{ mm}) \leq 35 \text{ } \mu\text{m}$ of form and sent the shell through the atmosphere at 8–9 km/s. **The heating was never calculated.** At 1,200 K the expansion of 176 mm of steel is 2,534 μm : **72 times the budget**. And both standard remedies make it worse — an **ablative** shield is consumed **by changing shape**, and a tiled shield adds joints that kill $v \geq L$ and bring down GATE-0.

It was resolved without materials, by two observations:

1. **The criterion is not ΔT : it is PERMANENT deformation.** Thermal expansion is reversible and the destination floor is established *after* cooling, so the only thing that breaks the floor is creep. For steel that is $T \leq 450 \text{ }^\circ\text{C}$, not 35 μm .
2. **Velocity and density can never both be high.** Radiative equilibrium at 450 $^\circ\text{C}$ gives a 12.4 kW/m² ceiling, and Sutton-Graves turns that into a maximum velocity per altitude: 0.42 km/s at sea level, 3.08 at 85 km, 4.80 at 100 km.

The 3.0 km/s that London–Berlin needs at 1 g therefore require being above **85 km**. The price is **4.4 minutes** of ascent and descent. **Every act is exo-atmospheric**, and the profile is designed with a heat-flux ceiling rather than a velocity ceiling. It is the max-Q management of any launch vehicle: a solved problem we had not looked at.

5-ter.3 The powers, and the door

The powers are separated: **A builds the act, B judges it**. B confirms **twice** — reservation and signature — and at commit **waives its vote**. The occupant turned out to be the **third vote**: a veto of their own up to commit, none after. A and B must be **independent signatories**, or the four eyes are two eyes twice.

only A	only B
propose, weigh, inspect, board	reserve or refuse without stating cause
seal (requires a live reservation from B)	verify the signature (second veto)
establish the origin floor	brake and receive
read and sign the message	establish its floor
pressurise · launch	apply the six criteria
abort below 30 % of route	OPEN THE DOOR — A cannot prevent it

Neither may open the blinding (the system does, with the sealed seed), abort above 30 %, or annul an act — the register annuls it, not a will. **Reconciliation is the only joint action.**

5-ter.4 The hard limits that appeared on costing

- **The bubble does not exceed ~5.03 m** (4–6 seats). With $\alpha = 1$ the field's disagreement grows linearly and the detection floor does not: they cross there. It does not matter, because $E \propto L^3 \propto N$ makes **energy per person flat** — there is no economy of scale to lose by splitting.

- **Stations may differ.** Dimension enters no formula of the contract, and B's floor may be up to $7.8\times$ worse than A's. Route asymmetry is **paid in time, not in g**.
- **Real cadence is 8.2 acts/day**, not 18.7: nobody had counted who cleans the cabin.
- **Fatigue is not the limit.** Retirement at 50,000 cycles (8.4 years), inspection every 5,000 (10 months), bounded by the six-month calendar already declared.

5-ter.5 Two advantages nobody had written, and the barrier nobody had seen

Radiation dose scales with time, and the act is 50–100× faster. A Hohmann transfer to Mars is 468 mSv — **78 % of an astronaut's career limit, in one trip** — against **6.3 mSv** for the act. It is the only quantity in the framework where haste **gives** safety. And the profile accelerates continuously, so **there is no microgravity**: artificial gravity for free, the whole way.

The barrier is elsewhere, and it is economic. The insurance premium does not need a measured rate: it needs a **bound**, and the same rule of three the framework uses for the FAR provides one. With the 37 reference acts, $P \leq 3/37 = 8.1\%$, and against a statistical value of life of €4 M the minimum premium is **€324,000 per act: 270 times the tariff**.

The framework is uninsurable today — not because it is dangerous, but because it is unknown. For the premium to fall to 10 % of the tariff requires ~112,000 acts without a single loss. It is the hardest barrier in the programme, and it can only be torn down by operating.

5-quater. THE ACT AS A SERVICE

5-quater.1 Who operates it

The four-eyes rule requires **two signatories per station**, and not any two: **a metrologist**, who signs the functional verification of the thirteen instruments and judges R^2 , α and the floor, and **an operations lead**, who signs weighing, inspection, boarding, sealing and commit.

Four people per act, two per station. It cannot go down to three: removing one breaks the four eyes, and *A builds / B judges* forbids the same person covering both stations even if they were alone. Nor is there one more: **there is no pilot** — nobody flies the capsule — there is no cabin crew, and there is **no blinding operator**, because the system opens it. The physician does not attend each act: fitness is certified by a **validity window**, as with an aeronautical medical certificate.

Of the 24 pipeline states, **only four admit a human decision**: reservation, sealing, signature verification and commit. Everything else is machine, procedure, or **forbidden to decide**.

The panel is not flown: it is witnessed. A Falcon 9 carries 20–30 people in the control room because there are decisions in flight. Here there are none: L4 has no controller. The framework's worst safety defect is its greatest staffing advantage.

Maintenance is not a team: cleaning and inter-act inspection are done by the operating shift, the periodic work is an external contract of ~194 h/year — external for the same reason the signatories are independent — and the in-house part is half a post for software, sealed registry and keys. **0.4 FTE**.

But **staffing is set by shift coverage, not by the act**: two posts $\times$ 5,840 h/year is **14.2 FTE** per station pair. **The act needs 4 people; the service needs 15.**

5-quater.2 The price, and the three scales that order it

Reconstructed tariff: **€1,202 per act at 1 g** (energy 362 + ascent 66 + empty return 190 + tooling 342 + station 23 + personnel 220). What orders everything is which block scales with what: **energy, ascent and return (€618) go with mass** and do not dilute; **tooling (€342) with shell area**; **station and personnel (€243) are per act** and divide whole among occupants.

seats	L cabin	price/person	g	transit	price
1	3.00 m	€1,203	0.50	14.5 m	€927
2	3.78 m	€1,011	1.00	10.3 m	€1,203
4	4.76 m	€894	2.00	7.3 m	€1,908
6	5.45 m !	€847	9.00	3.4 m	€6,353

No price per person exists below €618: that is the energy. The 5.03 m ceiling cuts at 4–6 seats, exactly where the curve stops falling steeply. Each g multiplies the price almost linearly — but lowering it has a new floor: the sealed leg is 6 min plus transit, and the CO₂ budget without active support is 15.8 min, hence **a ≥ 1.10 g without ECLSS**.

1 g remains the right point: the only value at which the occupant needs nothing.

5-quater.3 Where the framework stops coming last

payload	€/kg payload
10 kg	114
90 kg	13
146 kg (max., 3 m cabin)	8
600 kg	3

The shell weighs 888 kg and the passenger 90: 91 % of the energy moves the packaging. Against express air freight (€3–6/kg) we are at 2–3×, versus 5–8× in passenger terms.

The act is a better business moving things than moving people, because things need none of the €584 of fixed room cost, no medical certificate, no consent, and no CO₂ clock. It is the only market where the framework does not come last — and it finds it by giving up the thing that gave it its name.

By distance, price and airfare both grow linearly in d, the airfare from lower down: the ratio stays at 5–10× **and never improves**. What does change is time: to Sydney, 25.5 h against 2.6. **Time is what is sold, not price.**

6. What is measured and what is not

This is the section the framework exists to make possible, and the one that is least flattering.

Nothing in this document is a bench measurement of our own. Every figure is a derivation, a simulation, or a citation. The evidential scale is explicit:

doubt = $\Pi (1 - w_i)$ over INDEPENDENT lines, capped at 0.99
 MEAS_ADJ 0.95 · MEAS_PRO 0.85 · DER 0.70 · LIT 0.60 · DEC 0.30

Across the eight axes of completeness:

axis	
1 statement · 2 units · 3 evidential · 4 falsifiable · 5 costed · 6 window · 7 RESISTED	100 %
8 · MEASURED — there is a bench measurement, not a derivation or a citation	0 %
FRAMEWORK	87.5 %

The eighth axis was added **because** the first seven reached 100 %. An instrument that scores 100 % on everything **has stopped discriminating**, and the grades are set by whoever does the work: **a self-graded 100 % is worth less than a 98.7 % with named gaps**. The seven axes ask *is it well posed?* None asked *is it measured?*

The missing 12.5 points are a single axis costing €1,674:

test	cost	what it buys
GATE-0 with ring plate	€25	$\alpha \geq 0.95 \cdot D(176 \text{ mm}) \leq 35 \mu\text{m} \cdot v \geq L$
B1–B4 · re-placement floor	€80	10-re-placement floor · PCT_INV(n) · Shewhart
Complete act A→B	€534	Ω of the act · band A · FAR with its k · the six criteria
Traceability (certified standard)	€620	everything above moves from MEAS_PRO to MEAS_ADJ
Gap closure (desk and room)	€415	limbos, hot disconnect, A/B powers

GATE-0 is not the first test: it is the only one that can save the others. If $\alpha < 0.35$ the whole framework falls and the remaining €1,649 need not be spent.

7. Conclusions

1. A framework was built that is **completely posed, dimensioned, costed and resisted**, and that is **not measured**. Both halves of that sentence are the result.
2. The load-bearing claims are metrological, not physical. The programme proposes no new mechanism of nature, and where it is worth anything outside its own premise, it is worth it as method: $v \geq L$ as an acceptance criterion, GATE-0 as a correctability veto, and the **limbo catalogue** as a safety-engineering tool.
3. Several of the framework's own constraints turned out to be decisions rather than constraints. **9 g was the clearest case:** it bought seven minutes that the paperwork absorbs and cost $\times 9$ in energy, $\times 27$ in heat flux, /9 in reach, and seven of the thirteen occupant gaps.
4. The control flow, not the physics, held the largest gaps. Nine of seventeen abort branches had no destination, all of them with a person sealed inside, and the resolution — **the door is never the punishment** — had been true all along and unwritten precisely because it was true. **What is implicit cannot be audited.**
5. The economics are worse than the kinematics and the kinematics are better than expected. The act beats the aeroplane door to door at any distance and beats the train beyond 200 km; it is 5–8 $\times$ the price in passenger terms and 2–3 $\times$ **in freight**; and it is **uninsurable at today's evidence level by a factor of 270**.
6. The single most valuable action available to this programme costs **€25**.

8. References

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ANNEX A · COMPLETE FORMULARY

Forty-six formulas in eight blocks. Each carries its domain, units and evidential status: **MEAS** measured · **DER** derived · **LIT** literature · **DEC** declared.

I · CONTRACT

1. **Disagreement.** $\Omega = 1 - \text{corr}$ · dimensionless · **MEAS**.
2. **Band.** $A = \sqrt{(\Omega_{\text{imp}}/\Omega_{\text{act}})}$, against the **nearest** impostor · **MEAS**.
3. **Dead zone.** $dA/A = \frac{1}{2}\sqrt{(u_{\text{imp}}^2 + u_{\text{act}}^2)}$. With $u = 10 \%$, $A \in [0.934, 1.071]$ is a **tie, not a contract** · **DER**.
4. **False acceptance.** $\text{FAR} \leq 3/k$. A **FAR without its k says nothing**. Measured $\text{FAR} \leq 10^{-6}$ with $k = 3 \cdot 10^6$ · **MEAS**.
5. **Decidability index.** $d' = |\mu_I - \mu_G| / \sqrt{((\sigma_I^2 + \sigma_G^2)/2)}$ · **LIT** (Daugman). Measured in our act: $d' = 5.5$.

6. Multiplicity — best of k. $F_k(x) = 1 - [1 - F_0(x)]^k$. Effective multiplicity is **measured** by counting how many compete within the threshold; neither 1 nor $n!$ · **DER + LIT**.

7. Band law. $A = K_d \sqrt{a/\sigma}$, $K_2 = 0.632$, $K_3 = 0.597$. Robust to its own exponent · **DER + MEAS**.

8. Instantaneous—certified exclusion. $\psi = g K^4/A^4$, $g = 3 + (3 - \dim \text{ of symmetry group})$. In our act $\psi = 1.03 \cdot 10^{-4}$: the act is **9,713 times more material than informational** · **DER**.

II · INSTRUMENT

9. Dynamic threshold. $\delta = K_\Delta \cdot (\sigma_{\text{floor}}/\mu_{\text{floor}}) \cdot \sqrt{2}$, $K_\Delta = 3$; threshold = $(1 + \delta) \cdot \text{floor}$. The $\sqrt{2}$ is because there are two independent measurements · **DER + MEAS**.

10. Floor noise. relative error of $\delta = 1/\sqrt{2(n-1)}$. $n = 10 \rightarrow 24\%$; $n = 50 \rightarrow 10\%$. **The floor is measured by RE-PLACING, not by re-photographing** · **DER + MEAS**.

11. Inventory percentile. $PCT_INV = 100(1 - 1/2n)$. A fixed percentile lies at small n · **DER**.

12. Minimum constraints. $n_{\min}: d \cdot n - d(d+1)/2$ · **DER**.

13. Detection floor. 1.0 mm at 90 % power · **MEAS**.

14. Structure function. $D(r) = C \cdot r^\alpha$. α is a **FITTED** parameter, so the units of C change with each fit: a fixed threshold on C is not dimensionally valid. The criterion is on $D(r_0)$ in micrometres · **MEAS**.

15. Bridge to the Gaussian kernel. $D^2(r) = 2d[C(0) - C(r)]$; Gaussian kernel $\rightarrow \alpha = 1$, exponential $\rightarrow \alpha = 0.5$. $\alpha \geq 0.95$ requires a mean-square differentiable field · **DER + LIT**.

16. GATE-0 decision rule. VETO if $\alpha < 0.35$ (98.7 % reliable) · PASS if $\alpha \geq 0.95$ and $D(176 \text{ mm}) \leq 35 \mu\text{m}$ · INTERMEDIATE otherwise. Estimator: **Theil-Sen** · **MEAS**.

17. Correlation requirement — the law of the map. $v \geq L$, $v/L \rightarrow 0.6$. **Both shells must be formed in ONE operation with the same tooling** · **DER**.

III · MACHINE

Illumination of the reader · thermal reference of the field · **lithium-free energy in the sealed leg** · waste and fluids · communication with the outside · occupant instrumentation · propulsion interface · attitude control · $v_{\max} > 12 \text{ km/s} \Rightarrow \text{exo-atmospheric}$ · thermal soak in T(m) and at destination · the cabin is stored **in the room** · station energy · penetrations, how many and for what · two-face thermal sensor · motorised plate rotation · BIBS in smoke · inter-station data channel. Each carries its own threshold in the repository; all are **DER** or **DEC** pending bench.

IV · CUSTODY AND STRUCTURE

Wall thickness: steel $\geq 4 \text{ mm}$, aluminium ≥ 6.5 , composite ≥ 7.5 for a 3 m cabin · shell **monolithic**, imposed by $v \geq L$ · $\Delta T \leq 50 \text{ }^\circ\text{C}$ between faces · $T \leq 450 \text{ }^\circ\text{C}$ (creep), the criterion that replaces ΔT for the certificate · heat-flux ceiling $q = K\sqrt{(\rho/R) \cdot v^3} \leq 12.4 \text{ kW/m}^2$ at $\varepsilon = 0.8$ · validity 6 months or 1 act (IMCA D 024) · retirement at 50,000 cycles, inspection every 5,000 · FMEA: vacuum collapse, cycling fatigue, corrosion, brittle fracture, joint failure, thermal shock, supports and anchorage, penetrations, life-support autonomy, entrapment and egress, medical event.

V · OCCUPANT

26. Impact tolerance (< 2 s). $g \sim t^{(-0.461)}$, $R^2 = 0.968$ (Eiband). 45 g @ 44 ms · 25 g @ 0.2 s · 10 g @ 1.6 s. Onset $\approx 500 \text{ G/s}$. **Exponent 0.46 = constant ENERGY, not impulse**. ASTM D3332 uses exponent 1: resolved by **scope** — Eiband governs the occupant, D3332 the inert cargo, and the lesser governs · **LIT**.

27. Sustained tolerance (> 2 s). 4.5 g standing · 9 g reclined with anti-g suit · 25 g liquid immersion. **Eiband does NOT apply to a transit of minutes · LIT.**

28. Transit time. $t = 2\sqrt{(d/a)}$, $a = g_{\text{sustained}} \cdot 9.81$. **Valid only while $v = \sqrt{(a \cdot d)}$ stays sub-escape:** 12,786 km at 1 g, 1,421 km at 9 g · **DER.**

Also: $ppCO_2 \leq 3$ mmHg hourly mean · $O_2 \leq 23.5$ % with alarm · deluge ≤ 3 s · non-synthetic, O_2 -compatible materials · **pressure ramp ≤ 1.0 psi/min**, set by the middle ear and not by the shell's 13.5 · **zero loose objects** (a 1.4 kg laptop crosses a 3 m cabin at 83 km/h with 371 J) · **minimum age 12** under the neck criterion at 9 g, void at 1 g · medical fitness signed by a physician, by validity window.

VI · LIMBOS

39–46. $L = [t_{\text{start}}, t_{\text{end}}]$ an interval with a guarantee suspended; $\text{risk}(L) = \text{duration} \cdot P(\text{event}) \cdot (1 - P(\text{mitigation}))$; $E = \sum \text{risk}(L_i)$ in risk-equivalent minutes, **an UPPER BOUND, since limbos overlap**; $e(L, x) = (\partial L / L) / (\partial x / x)$ the elasticity of a limbo with respect to route, load or time.

Measured over a 50-minute act: $E = 12.8$, of which **L2 contributes 39 % and L4 27 % — 66 % between them with only 27 % of the time.**

*The most dangerous limbo is the shortest one. Ordering by duration and ordering by danger are **inverse**. L4 (transit, 3.4 min) has **null mitigation**: $e(L4, m) = 0.00$.*

VII · SCALE

31. Certification difficulty. $\Omega \propto 1/L$ — a larger cabin is **easier** to certify; α decreases with L — a larger cabin makes GATE-0 **impossible**. The two are opposed and $v \geq L$ comes out of the crossing · **DER.**

Hard ceiling. With $\alpha = 1$ and $D(176 \text{ mm}) = 35 \mu\text{m}$, $D(r) = 0.199 \mu\text{m/mm}$ · r crosses the 1.0 mm detection floor at $L = 5.03 \text{ m} \rightarrow 4\text{--}6$ seats · **DER.**

32. Mass and energy. $t_{\text{wall}} \propto L$, $M_{\text{shell}} \propto L^3$, $E \propto M \propto L^3$. From 30 cm to 3 m: $9 \rightarrow 888 \text{ kg}$, **×99 of energy** · **DER.**

VIII · METHOD

33. Evidential scale. $\text{doubt} = \Pi(1 - w_i)$ over independent lines, capped at 0.99; MEAS_ADJ 0.95 · MEAS_PRO 0.85 · DER 0.70 · LIT 0.60 · DEC 0.30.

34–46. TRIADA and MATE as procedure; declaration of every inter-layer import; PCT_INV(n); Theil-Sen as the default robust estimator; the rule of three for every rate; and the eight axes of completeness.

ANNEX B · Reproducibility

Every figure in this paper is produced by a script in `experiments/lib/` or `core_engine/src/`, and every script carries its TRIADA in its docstring: **INVENTORY**, the **MATHEMATICS written before running** with its numerical predictions, and the **MATE** reuse. Predictions are recorded as P1/P2/P3 and the outcome — confirmed or refuted — is recorded against them.

The chronological register is `docs/decisions/decisions_registry.md`; the accumulated method rules are `experiments/lib/REGLAS.md`; the completeness instrument is `experiments/lib/auditoria_marco.py` and prints the eight axes on every run.

Refuting this framework costs €25. Build a five-ring fiducial plate, photograph two trays of the same batch, fit $D(r) = C \cdot r^\alpha$ with Theil-Sen, and read α . **If $\alpha < 0.35$, the architecture does not hold and nothing else in this document needs to be spent on.**